

Prompt: **FULL-STACK FLUTTER UPAY SIMULATOR**

Track 01 — Trust & Risk Intelligence

You are an expert **Flutter + Dart, Python FastAPI, Machine Learning, Fraud Detection, Graph Analytics, NLP, and FinTech Security Engineer.**

Build a complete hackathon-ready **upay-inspired MFS simulation application** focused exclusively on:

TRACK 01 — TRUST & RISK INTELLIGENCE

The core mission is:

Protect customers, merchants, agents, and the wider MFS ecosystem from fraud, scams, account takeover, money-mule activity, and emerging financial risks.

This is a **simulation/prototype only**.

Do NOT connect to real upay accounts, real financial systems, real payment gateways, real PINs, real OTPs, or real customer data.

CORE PRODUCT CONCEPT

The most important product principle is:

“Invisible Protection for Customers, Actionable Intelligence for Fraud Analysts.”

The customer should have a simple and normal payment experience.

The customer should NOT see:

- Risk Score
- ML probability
- Fraud probability
- SHAP values
- Technical risk factors
- Internal fraud labels
- Model decisions

The customer only sees the normal transaction flow.

Behind the scenes, every sensitive transaction goes through the Trust & Risk Intelligence Engine.

CUSTOMER FLOW

The primary Send Money flow must be:

Login

↓

Home

↓

Send Money

↓

Enter Receiver Number

↓

Enter Amount

↓

Enter PIN

↓

Send

↓

Loading / Processing

↓

Backend Risk Intelligence Check

↓

Transaction Decision

↓

Success / Verification / Temporary Hold

The customer should feel that the normal payment system is simply processing the transaction.

During the loading state show:

Processing your transaction securely...

Do NOT show:

Risk Score: 82

or:

AI detected fraud.

ADMIN FLOW

The Admin/Fraud Analyst sees the intelligence layer:

Transaction



Risk Engine



Risk Score



Risk Level



Risk Factors



Explainability



Fraud Intelligence



Admin Action

Admin can see:

- Risk Score
- Risk Level
- Suspicious indicators
- Transaction history
- Device information
- Location information
- Velocity
- Amount deviation
- New receiver
- Suspicious transaction patterns
- Graph relationships
- AI explanation
- Recommended action

Admin can take actions such as:

- Review
- Hold
- Block
- Monitor
- Flag Account
- Escalate

TECH STACK

Frontend

Flutter

Dart

Material 3

Provider

HTTP / Dio

fl_chart

Google Fonts

Backend

Python

FastAPI

Pydantic

Uvicorn

Machine Learning

Python

Pandas

NumPy

Scikit-learn

XGBoost

Random Forest

Isolation Forest

Joblib

Explainable AI

SHAP

Graph Intelligence

NetworkX

NLP

For future/demo scam-text intelligence:

Bangla NLP

Bangla-BERT or lightweight text classifier

Data Generation

Faker

Pandas

NumPy

Admin Dashboard

Prefer:

Flutter Web

Do NOT create a separate Streamlit application unless necessary.

The main product should remain a **Flutter ecosystem**.

PROJECT STRUCTURE

Create:

upay_trust_risk/

|

├─ flutter_app/

|

├─ backend/

|

├─ ml/

|

├─ docs/

|

├─ screenshots/

|

└─ README.md

```
|  
└─ .gitignore
```

Flutter:

flutter_app/

```
|  
└─ lib/  
  │ └─ main.dart  
  │  
  │ └─ core/  
  │   │ └─ constants/  
  │   │ └─ theme/  
  │   │ └─ routes/  
  │   │ └─ utils/  
  │  
  │ └─ models/  
  │   │ └─ user.dart  
  │   │ └─ transaction.dart  
  │   │ └─ wallet.dart  
  │   │ └─ risk_result.dart  
  │   │ └─ security_event.dart  
  │  
  │ └─ services/  
  │   │ └─ api_service.dart  
  │   │ └─ auth_service.dart  
  │   │ └─ transaction_service.dart  
  │   │ └─ risk_service.dart
```

```
| |
| |─ providers/
| |   └─ auth_provider.dart
| |   └─ wallet_provider.dart
| |   └─ transaction_provider.dart
| |
| |─ screens/
| |   └─ splash/
| |   └─ login/
| |   └─ home/
| |   └─ send_money/
| |   └─ cash_out/
| |   └─ add_money/
| |   └─ transactions/
| |   └─ security/
| |   └─ profile/
| |   └─ admin/
| |
| └─ widgets/
|   └─ balance_card.dart
|   └─ quick_action.dart
|   └─ transaction_tile.dart
|   └─ security_status.dart
|   └─ risk_badge.dart
|   └─ loading_overlay.dart
|
```

└─ pubspec.yaml

Backend:

backend/

|

└─ app/

| └─ main.py

| |

| └─ api/

| | └─ routes/

| | └─ health.py

| | └─ transactions.py

| | └─ risk.py

| | └─ admin.py

| |

| └─ models/

| | └─ transaction.py

| | └─ risk.py

| | └─ user.py

| |

| └─ services/

| | └─ risk_engine.py

| | └─ feature_engineering.py

| | └─ fraud_detector.py

| | └─ graph_analyzer.py

| | └─ explanation_engine.py

| |

```
| └─ data/
|   └─ transactions.csv
|
|─ requirements.txt
└─ README.md

ML:

ml/
|
|─ generate_dataset.py
|─ train_model.py
|─ evaluate_model.py
|─ feature_engineering.py
|─ model.pkl
└─ dataset.csv
```

DEVELOPMENT MUST BE DONE IN 4 PHASES

Do NOT attempt to build everything blindly at once.

Build and test each phase before moving to the next phase.

PHASE 1 — FLUTTER CUSTOMER APP FOUNDATION

Goal

Build a polished, functional customer-side Flutter MFS simulator.

At the end of Phase 1, the app should work WITHOUT the ML backend.

Use local mock/demo data.

1. Splash Screen

Create:

upay-inspired logo

Trust & Risk Intelligence

Then navigate to Login.

2. Login Screen

Create:

Mobile Number

PIN

Login

Continue as Demo User

Use simulated authentication.

Do not store real credentials.

3. Home Screen

Display:

Good Morning, User

Available Balance

₹25,450.00

Quick actions:

Send Money

Cash Out

Add Money

Pay Bill

Recent transactions.

Also show:

Security Status

Protected

Your account is being monitored for suspicious activity.

Do NOT show risk scores.

4. SEND MONEY

Create the main customer flow:

Screen 1

Send Money

Receiver Number

Screen 2

Amount

₪_____

Screen 3

Enter PIN

Screen 4

Review

Receiver

Amount

Fee

Total

[Send]

Screen 5

After Send:

Processing securely...

Use a beautiful loading animation.

Screen 6

Money Sent Successfully

₪500 sent successfully.

Done

The user should NEVER see the internal risk analysis.

5. TRANSACTION HISTORY

Display:

Recent

Today

This Week

Each item:

Receiver

Amount

Date

Status

Customer can see normal transaction information.

Do NOT show:

Risk Score

Fraud Probability

ML Decision

6. CUSTOMER SECURITY CENTER

Show only user-friendly security information:

Security Center

Account Protected

Trusted Device

✓

Recent Login

✓

Security Monitoring

Active

Avoid exposing technical risk information.

PHASE 1 COMPLETION CRITERIA

Before proceeding:

✓ Flutter runs

✓ Login works

✓ Home works

✓ Balance works

- ✓ Send Money works
 - ✓ PIN flow works
 - ✓ Loading state works
 - ✓ Transaction history works
 - ✓ UI is responsive
 - ✓ No risk score shown to customer
-

PHASE 2 — FASTAPI + TRUST & RISK ENGINE

Goal

Connect the Flutter application to a real backend risk engine.

1. FASTAPI

Create:

GET /api/v1/health

POST /api/v1/predict-risk

POST /api/v1/transactions

GET /api/v1/transactions/{user_id}

GET /api/v1/admin/risk-events

GET /api/v1/admin/transactions

2. TRANSACTION REQUEST

Flutter sends:

```
{  
  "user_id": "USER001",  
  "receiver_id": "USER245",  
  "amount": 5000,
```

```
"transaction_type": "SEND_MONEY",  
"device_id": "DEVICE001",  
"location": "Dhaka",  
"timestamp": "2026-10-03T12:00:00"  
}
```

The PIN must NOT be sent or stored as plain text in the risk engine.

3. FEATURE ENGINEERING

Generate:

velocity

amount_deviation

time_anomaly

device_change

location_change

new_receiver

receiver_age

account_age

rapid_transaction

failed_attempts

Example:

velocity = transactions in last 1 hour

amount_deviation =

current_amount - user's normal amount

device_change =

new device compared to trusted device

location_change =

unusual location compared with history

4. MACHINE LEARNING

Implement:

Primary model

XGBoost

Fallback:

Random Forest

Use the model to generate a normalized:

Risk Score: 0–100

5. ANOMALY DETECTION

Implement:

Isolation Forest

Use it to identify unusual behaviour such as:

Large transaction

Unusual hour

Unusual location

New device

Rapid transfers

Combine anomaly information with the main model.

6. RISK CLASSIFICATION

Internal classification:

0–30

LOW

31–70

MEDIUM

71–100

HIGH

This classification is for the backend/admin only.

The customer does not see it.

7. POLICY ENGINE

Create:

ALLOW

VERIFY

HOLD

BLOCK

Example:

LOW

→ ALLOW

MEDIUM

→ VERIFY / HOLD

HIGH

→ HOLD / BLOCK

The final action must depend on configurable policy rules.

Do not hard-code the idea that every medium-risk transaction must always be blocked.

8. CUSTOMER EXPERIENCE

The customer continues to see:

Processing securely...

Then:

Safe

Money Sent Successfully

Suspicious requiring verification

Additional verification is required
to complete this transaction.

High-risk hold/block

Use neutral language:

Transaction temporarily unavailable.

For your security, this transaction
needs additional review.

Do NOT expose:

Risk Score: 91

Fraud probability: 93%

PHASE 2 COMPLETION CRITERIA

✓ Flutter ↔ FastAPI connected

✓ Transaction API works

- ✓ Risk engine works
 - ✓ Feature engineering works
 - ✓ ML model works
 - ✓ Isolation Forest works
 - ✓ Risk score generated
 - ✓ Customer never sees risk score
 - ✓ Backend makes transaction decision
-

PHASE 3 — ADMIN FRAUD INTELLIGENCE

Goal

Build the most important Track 01 presentation component:

ADMIN / FRAUD ANALYST DASHBOARD

This is where the hidden intelligence becomes visible.

Use Flutter Web.

1. ADMIN LOGIN

Create a separate demo Admin login.

Roles:

Customer

Fraud Analyst

Admin

For the prototype, use simulated authentication.

2. ADMIN DASHBOARD

Display:

Trust & Risk Intelligence

Total Transactions

25,430

High Risk

42

Medium Risk

186

Blocked

31

Under Review

18

3. LIVE RISK FEED

Show transactions as they arrive.

Example:

TXN-10342

User: USER001

Amount: ₺50,000

Risk Score

94/100

HIGH RISK

Indicators:

New Device

Unusual Location

High Amount

Rapid Transactions

4. TRANSACTION DETAILS

Admin clicks a transaction.

Show:

Transaction ID

User

Receiver

Amount

Timestamp

Device

Location

Risk Score

Risk Level

Decision

Then:

Risk Factors

5. EXPLAINABLE AI

Implement SHAP.

Admin should see:

Why was this transaction flagged?

1. Amount significantly exceeds user's normal pattern.
2. New device detected.
3. Multiple transactions occurred within a short period.

Show:

- Feature importance
- SHAP contribution
- Risk factor chart

Keep explanations understandable.

6. ADMIN ACTIONS

Buttons:

Review

Hold

Block

Monitor

Flag Account

Escalate

When Admin selects an action:

Confirm Action

Store the action in the backend.

Example:

TXN-10342

Risk: 94

Action: HOLD

Analyst: ADMIN001

Time: 12:42 PM

7. FRAUD CASE MANAGEMENT

Create:

Fraud Cases

Each case:

Case ID

User

Transaction

Risk Score

Reason

Status

Assigned Analyst

Created Time

Statuses:

Open

Under Review

Confirmed Suspicious

Resolved

False Positive

Important:

Do not automatically label a user as a criminal.

Use:

Suspicious Activity

Potential Fraud

Under Review

8. SECURITY ANALYTICS

Create charts:

Risk Distribution

Transaction Volume

High-Risk Trend

Blocked Transactions

Risk Score Distribution

Use `fl_chart`.

9. DEVICE INTELLIGENCE

Admin can see:

Trusted Device

New Device

Device Change Frequency

Example:

USER001

Trusted Devices: 2

New Device: DEVICE009

Last Seen: 2 minutes ago

10. LOCATION INTELLIGENCE

Display:

Usual Location

Current Location

Location Change

Example:

Normal:

Dhaka

Current:

Chattogram

Status:

Unusual Location

Use simulated data.

PHASE 3 COMPLETION CRITERIA

- ✓ Admin login
- ✓ Admin dashboard
- ✓ Live transaction feed
- ✓ Risk score
- ✓ Risk factors
- ✓ SHAP explanation
- ✓ Transaction details
- ✓ Admin actions
- ✓ Fraud cases
- ✓ Charts
- ✓ Device intelligence

✓ Location intelligence

PHASE 4 — ADVANCED TRUST INTELLIGENCE

This phase differentiates the project from a simple fraud-detection demo.

1. GRAPH ANALYSIS

Use:

NetworkX

Create transaction graph:

User

↓

Receiver

↓

Receiver

↓

Destination

Detect suspicious structures.

Analyse:

Degree Centrality

Connected Components

Transaction Clusters

Circular Transactions

High-Connectivity Accounts

2. MONEY-MULE INTELLIGENCE

Example:

$A \rightarrow B$

$A \rightarrow C$

$A \rightarrow D$

$A \rightarrow E$

$B \rightarrow X$

$C \rightarrow X$

$D \rightarrow X$

$E \rightarrow X$

Admin dashboard should highlight:

Potential Suspicious Network

Do not automatically claim money laundering.

Use:

Potential Money-Mule Pattern

Requires Investigation

3. ACCOUNT TAKEOVER DETECTION

Detect combinations such as:

New Device

+

New Location

+

Password/PIN failures

+

Unusual transaction

Then increase internal risk.

Admin sees:

Potential Account Takeover Pattern

4. VELOCITY INTELLIGENCE

Track:

Transactions / 5 minutes

Transactions / 1 hour

Amount / 1 hour

Unique receivers / 1 hour

Example:

Normal:

2 transactions/hour

Current:

17 transactions/hour

Admin sees:

Abnormal Transaction Velocity

5. BANGLA SCAM NLP

Create a prototype scam-text detection module.

Example:

"upay থেকে বলছি আপনার PIN দিন"

Classify:

Potential Credential Scam

Other examples:

"আপনার account বন্ধ হয়ে যাবে, OTP দিন"

"আপনি পুরস্কার পেয়েছেন, verification fee পাঠান"

"আপনার PIN confirm করুন"

Admin can see:

Scam Intelligence

Categories:

Phishing

Credential Request

Impersonation

Prize Scam

Suspicious

Normal

Do not expose sensitive model output directly to customers.

6. TRUST INTELLIGENCE PROFILE

Admin can open a user profile:

USER001

Account Age

320 days

Transactions

842

Average Amount

₺780

Devices

2

Locations

3

Risk Events

7

Blocked Transactions

2

Show behavioural timeline.

7. ADMIN RISK INVESTIGATION VIEW

Create a single powerful screen:

Transaction Investigation	
User: USER001	
Amount: ₺50,000	
Risk Score: 94/100	
Risk Level: HIGH	
Risk Factors	

New Device	
Amount Deviation	
Velocity	
Location Change	
Transaction Timeline	
Device History	
Location History	
Network Connections	
[Review] [Hold] [Block]	

This should be the primary judge-facing screen.

DEMO MODE

Create three predefined scenarios.

Scenario 1 — NORMAL

Amount: ₦500

Trusted Device

Normal Location

Normal Time

Normal Velocity

Backend:

Low Risk

ALLOW

Customer:

Money Sent Successfully

Admin:

Risk Score: Low

Scenario 2 — SUSPICIOUS

Amount: ₺8,000

New Receiver

Unusual Time

Backend:

Medium Risk

VERIFY / HOLD

Customer:

Additional verification required.

Admin:

Risk Score: Medium

Scenario 3 — HIGH RISK

Amount: ₺50,000

New Device

New Location

Unusual Time

Rapid Transactions

Backend:

High Risk

HOLD / BLOCK

Customer:

Transaction temporarily unavailable.

For your security, additional review is required.

Admin:

Risk Score: 94

HIGH RISK

New Device

Location Change

High Amount

High Velocity

DATASET

Generate synthetic data.

Minimum:

5,000 transactions

Suggested distribution:

80% Normal

20% Fraud-like / Anomalous

Features:

transaction_id

user_id

receiver_id

amount

timestamp

hour

device_id
device_change
location
location_change
account_age
receiver_age
velocity
amount_deviation
new_receiver
rapid_transaction
failed_attempts
is_fraud

Never use real customer data.

MODEL PIPELINE

Implement:

Synthetic Data

↓

Data Cleaning

↓

Feature Engineering

↓

Train/Test Split

↓

XGBoost / Random Forest

↓

Fraud Probability

↓

Risk Score 0–100

↓

Isolation Forest

↓

Anomaly Signal

↓

Policy Engine

↓

Decision

RISK SCORE DESIGN

Create a transparent normalized score:

0–30 Low

31–70 Medium

71–100 High

Risk score should combine model output and relevant anomaly signals.

Do not present this score to customers.

API RESPONSE

Example:

```
{  
  "transaction_id": "TXN10342",  
  "risk_score": 94,  
  "risk_level": "HIGH",
```

```
"decision": "HOLD",  
"customer_message": "Transaction temporarily unavailable.",  
"risk_factors": [  
  {  
    "feature": "device_change",  
    "impact": 0.31,  
    "message": "New device detected"  
  },  
  {  
    "feature": "amount_deviation",  
    "impact": 0.27,  
    "message": "Amount is significantly above normal behaviour"  
  },  
  {  
    "feature": "velocity",  
    "impact": 0.21,  
    "message": "Multiple rapid transactions detected"  
  }  
]  
}
```

Important:

Flutter Customer UI should only use:

decision

customer_message

transaction status

Admin UI can use:

risk_score

risk_level

risk_factors

impact

device

location

velocity

UI DESIGN PRINCIPLES

Create a professional Bangladeshi fintech experience.

Use:

- clean white backgrounds
- modern cards
- rounded corners
- clear typography
- subtle shadows
- accessible buttons
- smooth animations
- responsive layouts

Customer app:

Simple

Trustworthy

Fast

Minimal

Admin dashboard:

Data-rich

Analytical

Professional

Investigation-focused

Do not make the customer interface look like a cybersecurity dashboard.

CUSTOMER LANGUAGE

Use simple language.

Instead of:

High fraud probability

use:

For your security, this transaction needs additional review.

Instead of:

AI blocked your transaction

use:

Transaction temporarily unavailable.

The customer should never be unnecessarily alarmed.

ADMIN LANGUAGE

Admin can see technical information:

Risk Score

Risk Level

Anomaly Score

Feature Impact

Device Change

Velocity

Location Change

RESPONSIBLE AI

The system must:

- Use synthetic data.
- Avoid declaring a person a criminal.
- Treat model predictions as risk signals.
- Allow human review.
- Record analyst actions.
- Support false-positive investigation.
- Provide explainable reasons.
- Avoid exposing unnecessary risk information to customers.

Core principle:

AI assists the fraud analyst; it does not replace human investigation.

ERROR HANDLING

Handle:

Backend unavailable

Network timeout

Invalid API response

ML model unavailable

Database unavailable

Invalid transaction

Never silently bypass the risk engine.

If the risk service is unavailable, use a controlled policy such as:

Transaction requires additional review.

Do not automatically assume the transaction is safe.

TESTING

Backend tests:

Normal transaction

Medium-risk transaction

High-risk transaction

Large amount

New device

New location

High velocity

New receiver

API failure

Flutter tests:

Login

Home

Send Money

Receiver input

Amount input

PIN

Loading

Success

Verification

Temporary hold

Transaction history

Admin dashboard

GIT DEVELOPMENT STRATEGY

Maintain continuous Git commits.

Do NOT upload the entire project as one final bulk commit.

Suggested commits:

feat: initialize flutter project

feat: add login flow

feat: add customer dashboard

feat: add send money flow

feat: add transaction history

feat: create fastapi backend

feat: add transaction API

feat: add synthetic dataset

feat: add risk model

feat: integrate flutter with risk API

feat: add admin dashboard

feat: add shap explanations

feat: add graph fraud intelligence

feat: add demo scenarios

docs: update project documentation

FINAL HACKATHON STORY

The presentation should tell this story:

A customer simply wants to send money.

Customer:

Receiver Number

↓

Amount

↓

PIN

↓

Send

The customer sees:

Processing securely...

Behind the scenes:

Transaction

↓

Trust & Risk Engine

↓

Behaviour Analysis

↓

ML Risk Model

↓

Anomaly Detection

↓

Graph Intelligence

↓

Risk Score

↓

Policy Decision

Meanwhile, the Fraud Analyst sees:

HIGH RISK

Risk Score: 94/100

New Device

Unusual Location

Abnormal Amount

Rapid Transactions

The analyst takes:

HOLD

The customer receives:

Transaction temporarily unavailable.

For your security, additional review is required.

The customer's money is protected.

FINAL PRODUCT MESSAGE

The entire application must communicate this idea:

“The customer doesn't need to understand the risk. The system understands the risk, and the fraud analyst acts on it.”

And the broader Track 01 mission is:

“Protecting customers, merchants, agents, and the MFS ecosystem through intelligent, explainable, and human-supervised trust and risk intelligence.”

FINAL DELIVERABLES

At the end of Phase 4, the project must contain:

- ✓ Flutter Customer App
- ✓ Flutter Web Admin Dashboard
- ✓ Send Money Flow

- ✓ PIN Flow
- ✓ Secure Loading Flow
- ✓ Transaction History
- ✓ FastAPI Backend
- ✓ Synthetic Dataset
- ✓ XGBoost / Random Forest
- ✓ Isolation Forest
- ✓ Feature Engineering
- ✓ Risk Score
- ✓ Policy Engine
- ✓ SHAP Explainability
- ✓ Fraud Analyst Dashboard
- ✓ Admin Actions
- ✓ Fraud Case Management
- ✓ Device Intelligence
- ✓ Location Intelligence
- ✓ Velocity Detection
- ✓ Graph / Money-Mule Intelligence
- ✓ Account Takeover Detection
- ✓ Bangla Scam NLP Prototype
- ✓ Demo Scenarios
- ✓ API Documentation
- ✓ Unit Tests
- ✓ README
- ✓ Architecture Diagram
- ✓ Git Commit Plan

DEVELOPMENT RULE

Complete **Phase 1**, test it, and only then continue to **Phase 2**.

Complete **Phase 2**, test Flutter ↔ FastAPI ↔ Risk Engine, and only then continue to **Phase 3**.

Complete **Phase 3**, test the Admin Intelligence Dashboard, and only then continue to **Phase 4**.

Do not sacrifice functionality for visual complexity.

The priority order is:

Trust & Risk Intelligence → Working Transaction Flow → AI Detection → Admin Action → Explainability → Graph Intelligence → UI Polish.

 Starter Code Kit-এর অন্তর্ভুক্ত ফাইলসমূহ:

1. **generate_synthetic_data.py (Python Synthetic Data Generator):**

- বাংলাদেশ ব্যাংকের MFS লেনদেন নীতি ও উপায়ের ক্যাশ-আউট সীমা বজায় রেখে ১০০০+ সিন্থেটিক ট্রানজেকশন ডেটা জেনারেট করার পাইথন স্ক্রিপ্ট।

2. **main_api.py (FastAPI + XGBoost + SHAP Engine):**

- রিয়েল-টাইম ট্রানজেকশন রিস্ক স্কোরিং এবং **Explainable AI (SHAP)** রেসপন্স জেনারেট করার জন্য কমপ্লিট ব্যাকএন্ড API।
- এটি প্রতিটি লেনদেনের ক্ষেত্রে ALLOW, CHALLENGE_OTP, অথবা BLOCK অ্যাকশন নির্ধারণ করে।

3. **fraud_api_service.dart (Flutter API Integration Service):**

- **Dart** দিয়ে লেখা সার্ভিস ক্লাস, যা আপনার Flutter upay অ্যাপ সিমুলেটরকে সরাসরি FastAPI ব্যাকএন্ডের সাথে যুক্ত করে।

4. **README.md (GitHub Documentation Template):**

- অফিশিয়াল নিয়ম অনুযায়ী গিটহাবের বাধ্যতামূলক ৯টি সেকশনসহ (Overview, Tech Stack, Setup, Env Vars, Test Cases, Data Assumptions) রেডিমেড গিটহাব টেমপ্লেট।

5. 🌀 ১. Brand Color Palette (UpayColors)

```
6. import 'dart:ui';
7.
8. class UpayColors {
9.   static const Color primaryBlue = Color(0xFF0A2540);      // upay Navy
   Blue
10.   static const Color accentYellow = Color(0xFFFFFB800);    // upay
   Signature Yellow
11.   static const Color bgLight = Color(0xFFFF4F6F9);        // App
   Background
12.
13.   // AI Risk Level Indicators
14.   static const Color riskLow = Color(0xFF2ECC71);          // Safe
   (Green)
15.   static const Color riskMedium = Color(0xFFFF39C12);      // Warning
   (Orange)
16.   static const Color riskHigh = Color(0xFFE74C3C);         // High Risk
   Block (Red)
17. }
```

18.

19. 🌀 ২. pubspec.yaml Dependencies

20. আপনার Flutter প্রজেক্টের pubspec.yaml ফাইলে এই প্যাকেজগুলো যোগ করুন:

```
21. dependencies:
22.   flutter:
23.     sdk: flutter
24.   google_fonts: ^6.1.0      # বাংলা ফন্টের জন্য (Hind Siliguri)
25.   animate_do: ^3.3.4       # রিএক্টিভ অ্যানিমেশন
26.   pininput: ^4.0.0          # PIN & OTP ইনপুট ফিল্ড
27.   http: ^1.2.0              # Python API-র সাথে কানেক্ট করার জন্য
28.   lucide_icons: ^0.257.0    # আইকন কিট
```

29.

30. 🏠 ৩. Complete Flutter UI Widget (upay_home_screen.dart)

31. এই কোডটি সরাসরি আপনার Flutter অ্যাপে lib/main.dart বা

lib/upay_home_screen.dart ফাইলে ব্যবহার করতে পারবেন:

```
32. import 'package:flutter/material';
33. import 'package:google_fonts/google_fonts.dart';
34.
35. void main() => runApp(const UpayApp());
36.
37. class UpayApp extends StatelessWidget {
38.   const UpayApp({super.key});
39.
40.   @override
41.   Widget build(BuildContext context) {
42.     return MaterialApp(
43.       debugShowCheckedModeBanner: false,
44.       title: 'upay AI Fraud Shield',
45.       theme: ThemeData(
46.         textTheme: GoogleFonts.hindSiliguriTextTheme(),
```

```

47. scaffoldBackgroundColor: const Color(0xFFFF4F6F9),
48. ),
49. home: const UpayHomeScreen(),
50. );
51. }
52. }
53.
54. class UpayHomeScreen extends StatefulWidget {
55.   const UpayHomeScreen({super.key});
56.
57.   @override
58.   State<UpayHomeScreen> createState() => _UpayHomeScreenState();
59. }
60.
61. class _UpayHomeScreenState extends State<UpayHomeScreen> {
62.   bool _isBalanceVisible = false;
63.   final double _balance = 12500.50;
64.
65.   @override
66.   Widget build(BuildContext context) {
67.     return Scaffold(
68.       appBar: AppBar(
69.         backgroundColor: const Color(0xFF0A2540),
70.         elevation: 0,
71.         title: Row(
72.           children: [
73.             const CircleAvatar(
74.               backgroundColor: Color(0xFFFFFB800),
75.               child: Icon(Icons.person, color: Color(0xFF0A2540)),
76.             ),
77.             const SizedBox(width: 12),
78.             Column(
79.               crossAxisAlignment: CrossAxisAlignment.start,
80.               children: [
81.                 Text('উপায় ইউজার', style:
GoogleFonts.hindSiliguri(fontSize: 14, color: Colors.white70)),
82.                 Text('01700000000', style: GoogleFonts.inter(fontSize:
12, color: Colors.white54)),
83.               ],
84.             ),
85.           ],
86.         ),
87.         actions: [
88.           IconButton(
89.             icon: const Icon(Icons.security, color:
Color(0xFFFFFB800)),
90.             onPressed: () => _showAIProtectionStatus(context),
91.           ),
92.         ],
93.       ),
94.       body: SingleChildScrollView(
95.         child: Column(
96.           children: [
97.             // Balance Card Header
98.             Container(
99.               padding: const EdgeInsets.all(20),

```

```

100.         decoration: const BoxDecoration(
101.             color: Color(0xFF0A2540),
102.             borderRadius: BorderRadius.vertical(bottom:
103.                 Radius.circular(24)),
104.             child: Container(
105.                 padding: const EdgeInsets.symmetric(horizontal: 16,
106.                     vertical: 12),
107.                 decoration: BoxDecoration(
108.                     color: Colors.white.withOpacity(0.12),
109.                     borderRadius: BorderRadius.circular(30),
110.                     border: Border.all(color: const
111.                         Color(0xFFFFFB80).withOpacity(0.5)),
112.                     child: Row(
113.                         mainAxisAlignment: MainAxisAlignment.spaceBetween,
114.                         children: [
115.                             Row(
116.                                 children: [
117.                                     const Icon(Icons.account_balance_wallet,
118.                                         color: Color(0xFFFFFB80)),
119.                                     const SizedBox(width: 10),
120.                                     Text(
121.                                         _isBalanceVisible ? 'ট $ _balance' : 'ব্যালেন্স
122.                                         দেখতে ট্যাপ করুন',
123.                                         style: GoogleFonts.hindSiliguri(color:
124.                                             Colors.white, fontSize: 16, fontWeight: FontWeight.bold),
125.                                         ),
126.                                     ],
127.                                 ),
128.                                 IconButton(
129.                                     icon: Icon(_isBalanceVisible ?
130.                                         Icons.visibility_off : Icons.visibility, color: Colors.white70),
131.                                     onPressed: () => setState(() =>
132.                                         _isBalanceVisible = !_isBalanceVisible),
133.                                 ),
134.                             ],
135.                         ),
136.                     ),
137.                     const SizedBox(height: 20),
138.                     // AI Fraud Shield Live Status Bar
139.                     Padding(
140.                         padding: const EdgeInsets.symmetric(horizontal: 16),
141.                         child: Container(
142.                             padding: const EdgeInsets.all(12),
143.                             decoration: BoxDecoration(
144.                                 color: const Color(0xFF2ECC71).withOpacity(0.1),
145.                                 borderRadius: BorderRadius.circular(12),
146.                                 border: Border.all(color: const Color(0xFF2ECC71)),
147.                             ),
148.                             child: Row(
149.                                 children: [

```

```

147.         const Icon(Icons.verified_user, color:
148.           Color(0xFF2ECC71)),
149.         const SizedBox(width: 10),
150.         Expanded(
151.           child: Text(
152.             'upay AI Fraud Shield সক্রিয় আছে (রিয়েল-টাইম
153.               সুরক্ষায় সুরক্ষিত)',
154.             style: GoogleFonts.hindSiliguri(fontSize: 13,
155.               color: const Color(0xFF1E293B), fontWeight: FontWeight.w600),
156.             ),
157.           ),
158.         ),
159.
160.         const SizedBox(height: 20),
161.
162.         // Services Grid
163.         Padding(
164.           padding: const EdgeInsets.symmetric(horizontal: 16),
165.           child: GridView.count(
166.             shrinkWrap: true,
167.             physics: const NeverScrollableScrollPhysics(),
168.             crossAxisCount: 4,
169.             crossAxisSpacing: 12,
170.             mainAxisSpacing: 16,
171.             children: [
172.               _buildServiceItem(context, Icons.send, 'সেন্ড মানি',
173.                 Colors.blue, () => _simulateSendMoney(context)),
174.               _buildServiceItem(context, Icons.money, 'ক্যাশ আউট',
175.                 Colors.orange, () {}),
176.               _buildServiceItem(context, Icons.add_card, 'অ্যাড
177.                 মানি', Colors.green, () {}),
178.               _buildServiceItem(context, Icons.receipt_long, 'বিল
179.                 পে', Colors.purple, () {}),
180.               _buildServiceItem(context, Icons.shopping_bag,
181.                 'মার্চেন্ট পে', Colors.teal, () {}),
182.               _buildServiceItem(context, Icons.phone_android,
183.                 'মোবাইল রিচার্জ', Colors.pink, () {}),
184.               _buildServiceItem(context, Icons.savings, 'সেভিংস',
185.                 Colors.amber, () {}),
186.               _buildServiceItem(context, Icons.grid_view, 'আরও',
187.                 Colors.grey, () {}),
188.             ],

```

```

189. Widget _buildServiceItem(BuildContext context, IconData icon, String
    title, Color color, VoidCallback onTap) {
190.     return InkWell(
191.         onTap: onTap,
192.         borderRadius: BorderRadius.circular(16),
193.         child: Column(
194.             mainAxisAlignment: MainAxisAlignment.center,
195.             children: [
196.                 Container(
197.                     padding: const EdgeInsets.all(12),
198.                     decoration: BoxDecoration(color: color.withOpacity(0.12),
199.                         shape: BoxShape.circle),
200.                     child: Icon(icon, color: color, size: 26),
201.                 ),
202.                 const SizedBox(height: 6),
203.                 Text(title, style: GoogleFonts.hindSiliguri(fontSize: 12,
204.                     fontWeight: FontWeight.w500), textAlign: TextAlign.center),
205.             ],
206.         );
207.
208. // Simulate Send Money with AI Fraud Interception
209. void _simulateSendMoney(BuildContext context) {
210.     showModalBottomSheet(
211.         context: context,
212.         isScrollControlled: true,
213.         shape: const RoundedRectangleBorder(borderRadius:
214.             BorderRadius.vertical(top: Radius.circular(20))),
215.         builder: (context) => Padding(
216.             padding: const EdgeInsets.all(20),
217.             child: Column(
218.                 mainAxisAlignment: MainAxisAlignment.min,
219.                 crossAxisAlignment: CrossAxisAlignment.start,
220.                 children: [
221.                     Text('সেল্ড মানি করুন', style:
222.                         GoogleFonts.hindSiliguri(fontSize: 18, fontWeight: FontWeight.bold)),
223.                     const SizedBox(height: 12),
224.                     const TextField(
225.                         decoration: InputDecoration(labelText: 'প্রাপকের নম্বর',
226.                             border: OutlineInputBorder()),
227.                         keyboardType: TextInputType.phone,
228.                     ),
229.                     const SizedBox(height: 12),
230.                     const TextField(
231.                         decoration: InputDecoration(labelText: 'টাকার পরিমাণ
232.                         (ট)', border: OutlineInputBorder()),
233.                         keyboardType: TextInputType.number,
234.                     ),
235.                     const SizedBox(height: 20),
236.                     ElevatedButton(
237.                         style: ElevatedButton.styleFrom(backgroundColor: const
238.                             Color(0xFF0A2540)),

```

```

237.         onPressed: () {
238.             Navigator.pop(context);
239.             _triggerHighRiskAlert(context);
240.         },
241.         child: Text('এগিয়ে যান', style:
242.             GoogleFonts.hindSiliguri(fontSize: 16, color: Colors.white)),
243.     ),
244. ],
245. ),
246. ),
247. );
248. }
249.
250. // High Risk Warning Dialog (XAI Modal for Judges Demo)
251. void _triggerHighRiskAlert(BuildContext context) {
252.     showDialog(
253.         context: context,
254.         barrierDismissible: false,
255.         builder: (context) => AlertDialog(
256.             shape: RoundedRectangleBorder(borderRadius:
257.                 BorderRadius.circular(16)),
258.             title: Row(
259.                 children: [
260.                     const Icon(Icons.gpp_maybe, color: Color(0xFFE74C3C),
261.                         size: 30),
262.                     const SizedBox(width: 8),
263.                     Text('নিরাপত্তা সতর্কবার্তা', style:
264.                         GoogleFonts.hindSiliguri(color: const Color(0xFFE74C3C), fontWeight:
265.                             FontWeight.bold)),
266.                 ],
267.             ),
268.             content: Column(
269.                 mainAxisAlignment: MainAxisAlignment.min,
270.                 crossAxisAlignment: CrossAxisAlignment.start,
271.                 children: [
272.                     Text(
273.                         'অস্বাভাবিক ডিভাইস ও নতুন লোকেশন থেকে উচ্চ অঙ্কের লেনদেনের
274.                         চেষ্টার কারণে এই ট্রানজেকশনটি থামানো হয়েছে।',
275.                         style: GoogleFonts.hindSiliguri(fontSize: 14),
276.                     ),
277.                     const SizedBox(height: 12),
278.                     Container(
279.                         padding: const EdgeInsets.all(10),
280.                         decoration: BoxDecoration(color: Colors.red.shade50,
281.                             borderRadius: BorderRadius.circular(8)),
282.                         child: Column(
283.                             crossAxisAlignment: CrossAxisAlignment.start,
284.                             children: [
285.                                 Text('AI চিহ্নিত ঝুঁকির কারণসমূহ (SHAP Explanation):',
286.                                     style: GoogleFonts.hindSiliguri(fontSize: 12, fontWeight:
287.                                         FontWeight.bold, color: Colors.red.shade900)),
288.                                 const SizedBox(height: 4),
289.                                 Text('• Unrecognized Device Fingerprint (Weight:
290.                                     45%)', style: GoogleFonts.inter(fontSize: 11)),

```

```

282.             Text('• Velocity Surge: 3rd transaction in 2 mins
                (Weight: 30%)', style: GoogleFonts.inter(fontSize: 11)),
283.         ],
284.     ),
285. ),
286. ],
287. ),
288.     actions: [
289.         TextButton(
290.             onPressed: () => Navigator.pop(context),
291.             child: Text('বাতিল করুন', style:
                GoogleFonts.hindSiliguri(color: Colors.grey)),
292.         ),
293.         ElevatedButton(
294.             style: ElevatedButton.styleFrom(background-color: const
                Color(0xFF0A2540)),
295.             onPressed: () => Navigator.pop(context),
296.             child: Text('বায়োমেট্রিক দিয়ে যাচাই করুন', style:
                GoogleFonts.hindSiliguri(color: Colors.white)),
297.         ),
298.     ],
299. ),
300. );
301. }
302.
303. void _showAIProtectionStatus(BuildContext context) {
304.     showModalBottomSheet(
305.         context: context,
306.         builder: (context) => Container(
307.             padding: const EdgeInsets.all(20),
308.             child: Column(
309.                 mainAxisAlignment: MainAxisAlignment.min,
310.                 children: [
311.                     const Icon(Icons.shield, color: Color(0xFFFFB800), size:
                        50),
312.                     const SizedBox(height: 10),
313.                     Text('upay Real-Time Risk Shield', style:
                        GoogleFonts.inter(fontSize: 18, fontWeight: FontWeight.bold)),
314.                     const SizedBox(height: 6),
315.                     Text('আপনার অ্যাকাউন্টটি AI চালিত ফ্রড প্রোটেকশন ইঞ্জিনের সাথে
                        সংযুক্ত আছে।', textAlign: TextAlign.center, style:
                        GoogleFonts.hindSiliguri(color: Colors.grey.shade700)),
316.                 ],
317.             ),
318.         ),
319.     );
320. }
321. }

```